LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

DEPARTMENT OF ECONOMICS

Seminar: The Causes and Consequences of Migration

Summer Term 2026

Migrant Networks and International Trade

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Date of submission: 1 June 2026

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1 Introduction

The relationship between migration and international trade has long been central to international economics. Both flows are shaped by the same frictions of information, distance, and trust: bilateral trade depends not only on tariffs and transport costs but also on the search costs of finding counter-parties, the difficulty of enforcing contracts across legal systems, and the ability of buyers and sellers to communicate. Migrants are well placed to lower these frictions — by carrying knowledge of origin-country languages, regulations, and institutions, by retaining personal ties that substitute for weak contract enforcement, and by forming a demand base for origin-country varieties (Gould, 1994; Rauch, 2001; Rauch and Trindade, 2002). Whether these mechanisms generate a sizeable *causal* effect on trade is hard to establish, however, since migrants are not allocated across destinations at random and tend to settle in regions already internationally integrated.

This paper critically assesses two studies that confront this identification problem from opposite directions. Parsons and Vézina (2018) treat it as a natural-experiment problem and exploit the exodus of the Vietnamese Boat People to the United States after 1975: a quasi-random federal dispersal policy interacted with a roughly twenty-year US trade embargo on Vietnam, yielding a setting that closely approximates an experiment, from which they estimate a plausibly causal reduced-form pro-trade effect of migrant networks. Orefice et al. (2025), by contrast, treat it as a decomposition problem. Within a structural gravity framework they separate the bilateral migrant-network channel from two aggregate productivity channels — migration-induced knowledge diffusion and birthplace diversity — and find evidence that all three operate alongside one another.

Taken together, the two papers leave open how refugee-driven elasticities relate to broader migrant populations, whether short-run network formation can be observed, and whether formal subnational allocation rules can provide cleaner variation than historical settlement shares. The 2015/16 refugee inflow to Germany lets all three questions be reconsidered: protection seekers were distributed across the sixteen *Länder* under the *Königsteiner Schlüssel* (GWK, 2020), an administrative formula based on general state characteristics rather than origin-specific economic ties, and therefore a plausibly exogenous source of variation in local network size. Building on this setting, the second half of the paper develops a research proposal at the *Land*-origin level and presents preliminary estimates.

The remainder of the paper is structured as follows. Section 2 critically assesses the two assigned papers, with attention to their identifying assumptions and the open questions they leave. Section 3 introduces the research proposal. Sections 4–6 set out the empirical strategy, weaknesses, and data sources. Section 7 presents preliminary results, and Section 8 discusses what the design can and cannot deliver, relates the findings to other mechanisms of international economic integration, and concludes.

2 Summary and Critical Assessment of the Two Sources

The two assigned papers approach the migration-trade nexus from complementary methodological standpoints: Parsons and Vézina (2018) exploit a historical natural experiment to identify a clean total effect, while Orefice et al. (2025) develop a structural framework decomposing the effect into three channels. Together, they trace the central trade-off in this literature — between cleanly estimating *whether* migrants promote trade and understanding *how* they do so.

2.1 Parsons and Vézina: The Vietnamese Boat People as a Natural Experiment

2.1.1 Setting and theoretical motivation

Parsons and Vézina (2018) exploit two interlocking events. After the Fall of Saigon in 1975, a federal dispersal policy — explicitly designed to scatter refugees and "avoid another Miami" — resettled roughly 130,000 refugees across all fifty states through four military bases, with religious voluntary agencies placing most refugees by parish location in a process the authors describe as anarchic and driven by agency proactiveness rather than refugee choice. Baker and North (1984) document that only 47.3% were resettled in their stated state of choice, and that roughly 45% had moved states by 1980; the authors further show that the 1975 allocation is uncorrelated with state economic, political, or pre-existing Vietnamese characteristics. The US embargo on Vietnam, imposed in 1975 and lifted in February 1994, then created a roughly twenty-year window during which the shock unfolded while bilateral trade was zero, ruling out reverse causality.

Against this background, identification does not require a structural model. Parsons and Vézina do not develop one of their own; they motivate their reduced-form gravity specification with the network/search framework of Gould (1994) and Rauch and Trindade (2002). Within that framework, focusing on exports rather than imports isolates Gould's information channel — migrants lowering the cost of doing business with the origin — with networks expected to matter most for differentiated goods that lack the price-based signals of organised exchanges (Rauch, 2001).

2.1.2 Identification and results

Parsons and Vézina (2018) apply IV-PPML (Santos Silva and Tenreyro, 2006) to a cross-section of fifty US states and Washington, D.C. (51 observations), instrumenting the log 1995 Vietnamese stock — the first post-embargo year — with the log 1975 first-wave refugee count (first-stage correlation 0.98), alongside standard state-level controls and a 1995–2010 panel extension with state and year fixed effects.

A 10% increase in the Vietnamese network raises state exports to Vietnam by 4.5%–13.8% across specifications, and moving from the lowest to the highest state Vietnamese population raises 1995–2010 export growth by roughly 8 percentage points. Consistent with the network/search view, the effect is concentrated in differentiated goods (Rauch, 2001), with no effect on homogeneous or reference-priced goods. Placebos substituting ten other Asian destinations are uniformly insignificant, and a synthetic-control counterfactual implies top-ten host states would have exported around 50% less without their 1995 Vietnamese populations.

2.1.3 Critical assessment

However, several identification concerns merit emphasis.

Identification rests on observables

The balance tests cover observables only, and the dispersal policy was not a lottery: religious voluntary agencies placed refugees largely by parish geography, so unobserved state characteristics that made a state attractive to denominational resettlement infrastructure in 1975 — sponsorship density, civic-association ties, or the geography of dioceses — may not be fully independent of later Vietnam-specific trade orientation. The placebos likewise rule out only a generic Asian-network effect; they cannot rule out Vietnam-specific state characteristics, such as Pacific-Rim port infrastructure or NGO ties to Vietnam, correlated both with 1975 placement and with later Vietnam-specific trade. A second threat works in the opposite direction: refugee settlement itself may have reshaped state-level economic structures over the twenty-year window, opening a channel through which the 1975 instrument could affect 1995 exports beyond the migrant-stock channel. Card (1990) documents such reshaping for the Mariel boatlift.

Sample size and the "without California" robustness

Identification rests on 51 state-level observations, including Washington, D.C., with variation concentrated in a small set of high-Vietnamese states. The authors present specifications excluding California and the West Coast as evidence of robustness (Parsons and Vézina, 2018, Table 3, columns 3–4). However, the point estimate on the log Vietnamese stock more than doubles when California or the West Coast is excluded.[1] Survival of statistical significance under these exclusions is not the same as stability of point estimates: such movements suggest that inference should be interpreted cautiously, and that asymptotic inference for PPML is stretched at this sample size.

Cross-state spillovers

State-level estimation implicitly assumes that one state's Vietnamese network does not affect another state's exports. If cross-state spillovers exist (exporters serving multi-state buyers, family ties spanning state lines), this no-interference assumption would be violated, and the state-level treatment effect would be biased of unknown sign — toward zero if intermediation flows from agglomerated to dispersed states, or away from zero if it diverts trade across state lines.

[1] In Parsons and Vézina (2018), Table 3, the coefficient rises from 0.332*** in the baseline state-GDP-share specification to 0.677* without the West Coast and 0.766** without California.

Firm-level mechanism unidentified

Focusing on exports isolates Gould's (1994) information channel, but state-level aggregation cannot identify which firms drive the network effect, nor whether migrants are the relevant agents. Three competing mechanisms fit the results equally well — migrant entrepreneurship, migrant-supplied human capital within incumbent exporters, and a residual correlation in which migrants and exporters cluster for unrelated reasons — yet have different policy implications that state-level data cannot adjudicate.

2.2 Orefice, Rapoport and Santoni: Decomposing the Migration-Export Channel

2.2.1 Setting and theoretical motivation

Orefice et al. (2025) ask: of the total export-creating effect of immigration, how much operates through bilateral information networks — the channel Parsons and Vézina identify — and how much through aggregate productivity channels invisible to a country-pair design? They posit three mechanisms at different levels of aggregation.

The networks channel is bilateral: immigrants from origin j reduce informational and contracting frictions in trade with j. Knowledge diffusion (KD_{ikt}) is a country-sector spillover: immigrants from countries strong in sector k transfer productive know-how to destination firms, raising exports of k to *all* destinations; KD_{ikt} weights the bilateral immigrant stock by origin-revealed comparative advantage in k. Birthplace diversity (BD_{it}) is an aggregate productivity channel: a workforce drawn from many origins combines heterogeneous problem-solving skills, raising productivity disproportionately in cognitively complex sectors. BD_{it} is 1 minus the Herfindahl index of origin shares.

To address endogeneity, the IVs are micro-founded on the Random Utility Model derived in Appendix A, in which the unobservable component of utility follows a Type-1 Extreme Value distribution, yielding a multinomial-logit gravity equation for migration that supplies exogenous variation in bilateral stocks.

2.2.2 Identification and results

Orefice et al. (2025) identify the three channels in a two-step nested estimation on the structural gravity model (Head and Mayer, 2014). Step one identifies the networks channel from a PPML gravity regression of bilateral exports on the bilateral immigrant stock (country-pair-sector, 5-year windows, 1995–2015), with importer- and exporter-sector-year fixed effects absorbing multilateral resistance. Step two regresses the predicted exports from step one — a Costinot et al. (2012) proxy for revealed comparative advantage — on KD_{ikt} and BD_{it}.

Endogeneity is addressed through three IVs built on the RUM-implied gravity for migration, purged of destination-year fixed effects to remove endogenous local-labour-demand variation: 1960 origin shares as the historical share component, a Jaeger et al. (2018) correction removing a lagged-diaspora feedback term, and 1985–1990 origin-country natural disasters as an exogenous push shock. The results withstand additional robustness IVs and checks on the construction of KD_{ikt} and BD_{it}, the estimator, and the exclusion of major migration corridors.

The three main coefficients are significant at conventional levels — at the 1% level in nearly every estimation — and survive all alternative IVs. A 10% increase in the bilateral immigrant stock raises exports to the origin by 0.9–1.89% at the intensive margin, with significant extensive-margin effects. A 10% increase in KD_{ikt} raises country-sector exports by 1.56%, driven by transfers from high- to low-income economies. A 10% increase in BD_{it} raises exports by 1.1%, operating through the interaction with sector problem-solving intensity and concentrated in high-income exporters, where complex production processes account for a larger share of manufacturing.

2.2.3 Critical assessment

Orefice et al.'s identification chain is more ambitious than Parsons and Vézina's, and it rests on several assumptions worth inspecting.

Shift-share share-exogeneity

The shift-share IV combines 1960 bilateral migrant shares with national-level emigration shifts. Goldsmith-Pinkham et al. (2020) show that identification then rests on the exogeneity of those historical shares: destinations receiving many 1960 migrants from origin j must not differ systematically in ways that also drive their 1995–2015 trade with j. The 35-year lag aims to break such a correlation but cannot guarantee it, since pre-existing ties between country pairs may have both attracted 1960 migrants and continue to shape trade. Under the complementary shock-exogeneity framing of Borusyak et al. (2022), the authors pass a pre-trend test with no significant correlation between pre-1995 exports and post-1995 changes in the instrument, and their top-corridor exclusion exercise leaves the elasticity stable — mitigating but not eliminating the concern.

Unobserved sectoral allocation of immigrants

Both KD_{ikt} and BD_{it} assume immigrants are distributed homogeneously across sectors at destination, which the authors identify as the paper's principal limitation: immigrants from origins with a comparative advantage in a given sector need not actually work in that sector at destination. A second gap compounds it: KD_{ikt} proxies migrants' productive knowledge by origin revealed comparative advantage, even though migrants are not a representative draw from their origin's workforce. A robustness exercise using French district-sector data partially addresses the destination-side concern but covers a single country and cross-section. The mismatch remains: the mechanism operates through *what* workers do in destination firms, while the measure relies only on *where* they come from.

Natural-disaster IV: LATE and exclusion restriction

Natural-disaster pushes are plausibly exogenous at origin, but the resulting IV identifies a local average treatment effect, namely the effect for the subset of migrants whose decisions are actually triggered by disasters. This subset need not be representative of migrants relevant for trade overall. Separately, pre-1990 disasters may have persistent economic consequences in origin countries that directly affect their subsequent trading capacity, providing a path from the instrument to trade that bypasses migration and violates the exclusion restriction.

Generated regressor concern in the two-step

A subtle concern arises in the two-step nested estimation: step one's estimated exporter-sector-year fixed effects are interpreted as revealed comparative advantage (Costinot et al., 2012) and used as the dependent variable in step two. Yet, if immigration itself helps shape a country's comparative advantage — as the knowledge-diffusion and birthplace-diversity channels would imply — then immigration may partly shape the variable used to measure its aggregate productivity effect. This does not invalidate the exercise, but it makes the magnitude of the second-step estimates harder to interpret cleanly.

2.3 Open questions and motivation for the research proposal

Three open questions motivate the research proposal in Section 4.

Refugee versus economic migration

Parsons and Vézina's identification depends on a refugee shock, while Orefice et al. pool migration types and partly use disaster-driven migration as an instrument. How elasticities from these specific margins relate to broader migrant populations remains open, given likely differences in skills, return migration, and attachment to the origin. This motivates a second refugee-based estimate in a different institutional context.

Short-run versus long-run evidence

The twenty-year US embargo on Vietnam forced an unusually long lag between the migration shock and the observable trade response, leaving open whether migrant networks affect exports at shorter horizons or require decades to build up. Jaeger et al. (2018) raise a related concern: when migration and trade evolve jointly, shift-share designs may confound short- and long-run effects. This motivates a setting in which the shock is observed at an annual frequency shortly after arrival.

Subnational variation under a formal allocation rule

Parsons and Vézina rely on parish-based refugee placement, while Orefice et al. rely on historical settlement shares. Both approaches leave room for concerns that the share component captures pre-existing economic or institutional ties. A formal subnational allocation rule can therefore provide cleaner variation if it is set independently of origin-specific migrant networks or trade relationships.

Other relevant mechanisms

A further open question is whether the trade response captured by both papers exhausts migration's effect on international economic integration: migration may also operate through diaspora-driven foreign direct investment (Burchardi et al., 2019), remittance-financed import demand, migrant entrepreneurship, and the return-migration knowledge-transfer channel documented for the German-Yugoslav case by Bahar et al. (2024). The proposed *Land* $\times$ origin design is silent on these margins by construction; Section 8 returns to firm- and worker-level extensions.

3 Research Proposal: Introduction

Did the 2015/16 refugee inflow into Germany raise *Länder* exports to the main origin countries of protection seekers? The question combines the refugee-shock logic of Parsons and Vézina (2018) with the panel shift-share structure of Orefice et al. (2025): exposure varies across *Länder* and origin countries, while annual trade data enable comparisons of pre- and post-shock export dynamics.

3.1 The natural experiment

Between 2014 and 2016, Germany experienced a large, sudden influx of protection seekers. Their geographic allocation across the sixteen *Länder* followed the *Königsteiner Schlüssel* (GWK, 2020), the federal allocation formula described in Section 4.2. The proposal focuses on Afghanistan, Eritrea, Iraq, Iran, and Syria, the five origins accounting for the bulk of the 2014–2016 increase in protection-seeker stocks. Because the key is based on general state characteristics rather than origin-specific economic ties, it provides plausibly exogenous variation in predicted *Land*-origin exposure.

3.2 Connection to the two sources

The design combines elements of both assigned papers. Like Parsons and Vézina (2018), it uses refugee allocation rather than voluntary migrant sorting to obtain plausibly exogenous variation in local migrant stocks. Unlike their US setting, however, the response is observed annually across German *Länder* shortly after the 2015/16 inflow rather than after a long embargo period. Like Orefice et al. (2025), the design follows a shift-share logic , allocating national-origin-specific inflows across destinations. Its key departure is the share component: a written administrative formula replacing historical settlement shares.

3.3 Related study

A closely related contribution is Bahar et al. (2024), who exploit the repatriation of around 700,000 Yugoslavian refugees who had received temporary *Duldung* protection in Germany in the early 1990s. Using a spatial dispersal instrument for asylum-seeker allocation across German regions and a difference-in-differences design on industry-level trade, they find that a 10% rise in industry-specific returning refugees raises home-country exports by 1% to 1.6%, with the effect driven by occupations most likely to transfer knowledge. While the German asylum-dispersal setting has been widely used to study refugee labour-market and regional outcomes, the trade margin remains comparatively unexplored: their setting answers the return-migration / knowledge-diffusion leg of Orefice et al.'s framework, while the present design asks the complementary destination-side question of whether refugees who remain in Germany raise origin-specific exports out of Germany.

3.4 Open questions addressed and expected contribution

The expected contribution follows from the three open questions raised in Section 2. First, the design delivers a refugee-based elasticity of migration-trade in a contemporary European setting, complementing Parsons and Vézina (2018) and Orefice et al. (2025). Second, it observes the response at an annual frequency within 9 years after the shock, addressing the short- versus long-run blending concern raised by Jaeger et al. (2018). Third, it replaces historical settlement shares with a formal administrative allocation formula, addressing the share-exogeneity concern of Goldsmith-Pinkham et al. (2020). It does not distinguish between refugee and economic-migrant elasticities or identify the firm-level mechanism — limitations taken up in Sections 5 and 8. Whether the design recovers a positive effect or, as Section 7 reports, an informative null is itself part of the contribution.

4 Proposed Empirical Strategy and Identifying Assumptions

4.1 Setting, sample, and unit of observation

The unit of observation is the *Land* $\times$ origin country $\times$ year cell, with all sixteen *Länder* observed annually. Origins are restricted to Afghanistan, Eritrea, Iraq, Iran, and Syria because these five origins combine a meaningful pre-shock period with sufficient observed exposure across *Länder* for the proposed panel design. The panel spans 2010–2025, partitioned into a pre-period (2010–2014), the shock period (2015–2016), and a post-period (2017–2025). The outcome is the export value from each *Land* to each origin, in thousand euros, taken from the official German federal-state foreign trade statistics. Working in levels rather than logs preserves zero flows, which are not negligible in this sample. After restriction to the analysis sample, the resulting unbalanced panel contains 1,247 cell-year observations across 80 *Land* $\times$ origin pairs.

4.2 Specification

Let X_{ijt} denote exports from *Land* i to origin j in year t and T_{ij} the actual stock of protection seekers from j in i at the end of 2016. Define the post-period indicator $\text{Post}_t \equiv \mathbf{1}\{t \geq 2017\}$. The preferred specification interacts T_{ij} with Post_t and estimates

$$X_{ijt} = \exp(\beta\, T_{ij} \cdot \text{Post}_t + \theta_{ij} + \theta_{it} + \theta_{jt})\, \varepsilon_{ijt},$$

where θ_{ij}, θ_{it} and θ_{jt} are *Land* $\times$ origin, *Land* $\times$ year, and origin $\times$ year fixed effects, and the treatment is rescaled per 1,000 persons. Since T_{ij} is time-invariant within a pair and would otherwise be absorbed by θ_{ij}, identification requires interacting it with Post_t. The pair fixed effect absorbs time-invariant bilateral determinants (distance, language, diaspora history);

θ_{it} absorbs all *Land*-year shocks (GDP, employment, manufacturing structure, total export capacity), which is why no regional controls enter the main specification and instead reappear only in a robustness check that drops θ_{it}; θ_{jt} absorbs all origin-year shocks (conflict intensity, exchange rates, sanctions). T_{ij} is endogenous because protection seekers settle in *Länder* for reasons that may correlate with bilateral trade potential. The excluded instrument is the *Königstein*-predicted stock,

$$\widehat{T}_{ij} \;=\; s_i^K \cdot N_j^{2016},$$

where s_i^K is the *Land*'s *Königstein* share averaged over 2015 and 2016, the years governing the cohort's allocation, and N_j^{2016} is the national protection-seeker stock from origin j in 2016. The *Königsteiner Schlüssel* allocates asylum seekers across the *Länder* according to a written formula based on population and tax revenue, with population weighted one third and tax revenue two thirds; it is calculated by the Joint Science Conference independently of origin-specific trade relationships (GWK, 2020). The instrument, like the treatment, is interacted with Post_t and rescaled per 1,000 persons.

The preferred estimator is PPML, motivated by the presence of zeros and the multiplicative form of the gravity equation (Santos Silva and Tenreyro, 2006; Head and Mayer, 2014). To check sensitivity to estimator choice, the analysis additionally implements a linear IV/2SLS specification on $\log(X_{ijt}+1)$ and a control-function IV-style PPML in which the first-stage residual is included as an additional regressor in the PPML second stage. A further robustness check replaces the 2016 national stock N_j^{2016} with the 2014–2016 change $N_j^{2016} - N_j^{2014}$ to verify that the conclusion does not depend on defining the shock as a level or a change. Standard errors are clustered by *Land* $\times$ origin throughout.

4.3 Identifying assumptions

Relevance

The IV strategy requires that predicted *Königstein* exposure correlates sufficiently strongly with actual exposure, conditional on the fixed effects. Relevance is assessed via the first-stage regression

$$T_{ij}\cdot\text{Post}_t \;=\; \pi\,\widehat{T}_{ij}\cdot\text{Post}_t \;+\; \theta_{ij} + \theta_{it} + \theta_{jt} \;+\; u_{ijt}.$$

Instrument strength is evaluated using the conventional first-stage F-statistic benchmark and further scrutinised by re-estimating the first stage on five leave-one-origin-out samples. A strong first stage supports relevance, but by itself, not the exclusion restriction.

Share-exogeneity

The share-exogeneity requirement (Goldsmith-Pinkham et al., 2020), scrutinised for Orefice et al. (2025) in Section 2.2.2, applies here too. Because the *Königsteiner Schlüssel* is set by a legal formula and its components — population and tax revenue — are predetermined relative to the 2015/16 wave, the shares satisfy this requirement more directly than the 1960 historical-settlement shares used in Orefice et al. (2025). The residual concern is that the same components may still correlate with general *Land* export capacity; identification, therefore, rests on the fixed effects and pre-trend test rather than the formula alone.

No differential pre-trends

Jaeger et al. (2018) stress that migration shift-share designs are vulnerable to spurious correlation when areas exposed to the shock were already on different trends. This is assessed with a Borusyak-Hull-Jaravel-style cross-sectional first-differences test (Borusyak et al., 2022), regressing the 2010–2014 pair-level change in $\log(X_{ijt}+1)$ on the later *Königstein*-predicted exposure assigned to each pair. The regression is cross-sectional at the pair-level with separate *Land* and origin fixed effects, since pair fixed effects would absorb the cross-sectional instrument. It tests for differential pre-shock trends across later high- and low-exposure pairs.

Exclusion restriction

The exclusion restriction requires that, conditional on the fixed effects described above, predicted *Königstein* exposure affects post-2016 origin-specific exports only through actual local exposure. The three-way FE structure absorbs the natural threats: federal compensation transfers and *Land*-level asylum or integration policies are absorbed by θ_{it}, since neither varies by origin within *Land*-year; direct origin-specific channels such as bilateral diplomacy or sanction relief are absorbed by θ_{jt}; and permanent *Land*-origin trade relationships are absorbed by θ_{ij}. What remains is a *Land* $\times$ origin $\times$ time-varying shock that correlates with predicted exposure but operates outside actual exposure. The pre-trends test and the event-study coefficients would pick up such a confounder if it is already operative before 2017, but not if it is contemporaneous with the wave.

5 Weaknesses of the Approach

Five limitations of the design should be noted.

Untested short-run horizon

The short post-shock horizon is the price of focusing on the immediate aftermath of the 2015/16 wave. Trade-network formation operates partly through firm entry and the discovery of new buyers, so any null finding within this design may also reflect that such effects have not yet materialised.

Secondary mobility after allocation

The *Königsteiner Schlüssel* governs initial allocation only, and secondary mobility after asylum is granted may weaken the link between predicted and actual exposure in later years. The short-horizon endpoint specification (2014–2017) in Appendix Table A1 and the event-study specification in Figure 1 can detect attenuation in the dynamic profile. Still, it cannot eliminate the underlying problem if secondary mobility is severe.

Mechanism unidentified at *Land*-level

The aggregation is too coarse to identify a mechanism — the same firm-level limitation raised for Parsons and Vézina in Section 2.1.3. *Land*-level data cannot reveal whether migrants are themselves exporters, are employed by exporting firms, or are merely correlated with them, so the design cannot test the micro-level information and trust channels described by Rauch (2001) and Rauch and Trindade (2002). Linked employer-employee or firm-level trade data, discussed in Section 8, would be the natural extension.

Small origin cross-section

The cross-sectional dimension on origins is small ($J = 5$), so identifying variation rests on a narrow base even though the panel contains 80 *Land*-origin pairs. The leave-one-origin-out exercise in Appendix Table A2 checks whether any single origin drives the result, but cannot expand the structural dimension of the cross-section.

Cross-*Land* spillovers

The design inherits the no-interference assumption flagged for Parsons and Vézina in Section 2.1.3: exposure in one *Land* is assumed not to affect another *Land*'s exports to the same origin. This is plausible for exports recorded at the *Land* of dispatch, but could fail for multi-*Land* buyers or cross-border supply chains. The leave-one-origin-out exercise in Appendix Table A2 probes influential-pair concerns but does not bound cross-*Land* spillovers directly.

6 Data Sources and Data Access

All variables in the main analysis are publicly available, free of charge, and require no formal application, reconstructed from open Destatis GENESIS-Online tables and the *Königsteiner Schlüssel* published by the Gemeinsame Wissenschaftskonferenz (GWK).[2]

Outcome and treatment

The outcome is *Land*-to-origin exports in thousand euros, from Destatis table 51000-0032 (*Außenhandel: Bundesländer*), with $\log(X_{ijt} + 1)$ used in the linear IV; the table also reports export weight in tons, used in one robustness check. The treatment is the bilateral stock of protection seekers from origin j in *Land* i, from table 12531-0024 (*Schutzsuchende*), aggregated to the *Land* $\times$ origin level for 2014 and 2016, and the implied change; the table is extended to 2017 for the delta-endpoint check.

[2] Standard bilateral gravity controls, such as distance, common language, regional trade agreements, and tariffs, are available from the CEPII Gravity Database (Conte et al., 2022) and correspond to the gravity controls discussed in Head and Mayer (2014). Since Germany is the only destination, these variables vary only by origin or origin-year and are absorbed by the *Land*-origin and origin-year fixed effects in the preferred specification. They are therefore archived for transparency but do not enter the estimating equations.

Instrument and controls

The *Königsteiner Schlüssel* is published annually by the GWK (GWK, 2020); the preferred basis is the 2015–2016 average, with the 2014 key and the 2014–2016 three-year average as robustness checks. A battery of *Land*-level controls enters only the robustness exercise that drops θ_{it} (Table 2, "Regional controls"), since the preferred specification absorbs them in θ_{it}.[3]

Access and limitations

The Destatis tables are available for download via GENESIS-Online, and the *Königsteiner Schlüssel* is published in the Bundesanzeiger and on the GWK website. The main limitation is that all variables are aggregated at the *Land*-level; identifying whether protection seekers themselves work in exporting firms would require restricted-access firm- or worker-level microdata, discussed as a follow-up in Section 8.

Replication

All code and the assembled data required to reproduce every table and figure in this paper are openly available and permanently archived at Zenodo (Chinamere, 2026).

7 Data Analysis and Results

This section presents preliminary estimates from the design described in Sections 4–6. The analysis is built around one preferred specification — the PPML reduced form — and surrounded by a first-stage test, a pre-trend check, an event study, and a set of robustness exercises whose role is to probe its credibility rather than to make independent claims. Given the short panel, the small cross-section of origins ($J = 5$), and the nine-year post-period, the results should be read as a first pass rather than as a final answer.

7.1 First stage and instrument relevance

Column (1) of Table 1 reports the first stage. A one-unit increase in *Königstein*-predicted exposure per 1,000 persons is associated with a 0.998-unit increase in the actual end-of-2016 protection-seeker stock per 1,000 persons (SE 0.206), with a first-stage F-statistic of 23.4. The coefficient is significant at the 1% level, and the F-statistic exceeds the conventional weak-instrument threshold of ten, supporting instrument relevance.

7.2 Preferred PPML reduced form

Column (2) reports the preferred PPML reduced form. The coefficient on the *Königstein*-predicted exposure interacted with the post-2016 indicator is -0.0001 (SE 0.0044), which is statistically indistinguishable from zero. Thus, an additional 1,000 predicted protection

[3]GDP is taken from GENESIS table 82111-0010 (*VGR der Länder, BIP*), population from 12411-0010 (*Bevölkerung: Bundesländer, Stichtag*), the unemployment rate from 13211-0007 (*Arbeitsmarktstatistik*), employment from 13311-0002 (*Länderberechnung Erwerbstätige*), and gross value added by economic sector from 82111-0011 (*VGR der Länder, Bruttowertschöpfung*), from which the manufacturing share is constructed. Total exports to the world are constructed by aggregating GENESIS table 51000-0032 across all destinations.

seekers from a given origin is associated with effectively no change in post-period exports to that origin. The preferred specification, therefore, yields no evidence of a positive export response to refugee-induced regional exposure at the *Land*-origin level.

7.3 Robustness of the estimator

The remaining columns of Table 1 show that the null is not specific to the preferred PPML reduced form. The non-instrumented PPML benchmark in column (3), the linear IV/2SLS specification using $\log(X_{ijt}+1)$ in column (4), and the control-function IV-style PPML in column (5) all yield small and statistically insignificant coefficients. The null pattern, therefore, survives the main alternatives to the preferred estimator.

Table 1: Main results

	(1) First stage	(2) PPML reduced form	(3) PPML benchmark	(4) Linear IV / 2SLS	(5) CF-IV PPML
IV: predicted exposure (/1,000)	0.998***	-0.0001			
	(0.206)	(0.0044)			
Treatment: actual exposure (/1,000)			0.0001	0.0064	-0.0003
			(0.0032)	(0.0072)	(0.0047)
First-stage residual (/1,000)					0.0009
					(0.0103)
Outcome	Treatment	Exports	Exports	log Exports	Exports
Estimator	feols	fepois	fepois	feols-IV	fepois + CF
Fixed effects:					
Land × origin	Yes	Yes	Yes	Yes	Yes
Land × year	Yes	Yes	Yes	Yes	Yes
Origin × year	Yes	Yes	Yes	Yes	Yes
First-stage F	23.4				
Observations	1,247	1,247	1,247	1,247	1,247
Clusters (pair)	80	80	80	80	80

Notes: Robust standard errors clustered at the *Land* × origin country level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The instrument is the *Königstein*-predicted protection-seeker stock in 2016, interacted with a post-2016 indicator, scaled per 1,000 persons. The treatment is the corresponding actual stock. Column (1) is the first-stage `feols` regression. Columns (2) and (3) are PPML specifications estimated via `fepois`. Column (4) reports $\log(X_{ijt}+1)$ as the outcome to handle zero flows. Column (5) implements a control-function IV-style PPML approach: the first-stage residual from a linear regression of treatment on the instrument is included as an additional regressor in the PPML second stage. Sample period: 2010–2025.

7.4 Pre-trend evidence

A central concern in any shift-share migration design is that exposure may correlate with pre-existing differences in outcome trends (Jaeger et al., 2018). Column (1) of Table 2 reports the Borusyak-Hull-Jaravel-style pre-trend test described in Section 4. Regressing the 2010–2014 change in $\log(X_{ijt}+1)$ at the pair-level on later *Königstein*-predicted exposure yields a coefficient of 0.0026 (SE 0.0175). While no pre-trend test is dispositive, this result is reassuring: pairs later assigned to high predicted exposure show no sign of being on differential pre-shock export trends.

7.5 Dynamic response: event study

Figure 1 reports the linear event-study estimates, with 2014 as the reference year and the 2015–2016 shock period shaded. The year-specific coefficients remain close to zero and statistically insignificant throughout, with no clear post-2016 increase in exports. The figure is therefore consistent with the formal pre-trend test in Table 2 and with the static null in Table 1. The PPML version is reported in Appendix Figure A1. It also shows no positive post-shock response, but is interpreted only as a descriptive robustness diagnostic because several pre-period coefficients are negative as well.

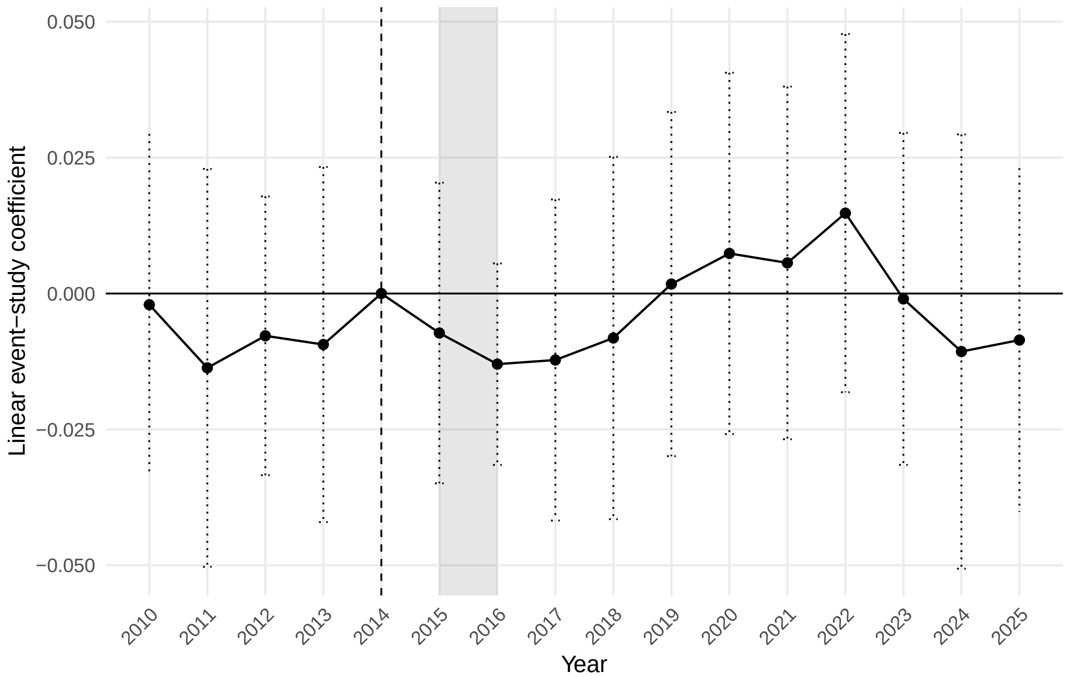

Figure 1: Linear event-study estimates

Notes: The figure plots year-specific coefficients on the interaction between year indicators and the 2016 *Königstein*-predicted exposure measure, with 2014 as the reference year. Dotted whiskers indicate 95% confidence intervals constructed from cluster-robust standard errors at the *Land* $\times$ origin level. The shaded band indicates the 2015–2016 shock period. The estimates provide no evidence of a differential post-shock export response.

7.6 Additional main specifications

Columns (2) and (3) of Table 2 report two additional main specifications. Replacing the 2016 stock exposure, with the 2014–2016 predicted delta exposure, yields a coefficient of -0.0001 (SE 0.0049). Replacing θ_{it} with observable regional controls yields -0.0109 (SE 0.0069). Neither coefficient is statistically significant, so the null is not driven by the stock definition or by the saturated *Land* $\times$ year fixed effects of the preferred specification.

Table 2: Pre-trend test and additional main specifications

	(1) BHJ pre-trend	(2) Delta PPML RF	(3) Regional controls
IV: predicted exposure (/1,000)	0.0026		-0.0109
	(0.0175)		(0.0069)
IV: predicted delta exposure (/1,000)		-0.0001	
		(0.0049)	
Outcome	Δlog Exports	Exports	Exports
Estimator	feols	fepois	fepois
Sample period	2010–2014	2010–2025	2010–2024
Fixed effects:			
Land × origin	No	Yes	Yes
Land × year	No	Yes	No
Origin × year	No	Yes	Yes
Land (separate)	Yes	-	-
Origin country (separate)	Yes	-	-
Regional controls	No	No	Yes
Observations	77	1,247	1,169
Clusters	16	80	80

Notes: Standard errors are reported in parentheses. Standard errors are clustered at the *Land* × origin country level in columns (2) and (3), and at the *Land*-level in column (1). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Column (1) is a Borusyak-Hull-Jaravel-style cross-sectional first-differences pre-trend test, regressing $\Delta\log(X_{ijt}+1)$ between 2010 and 2014 on the IV with separate *Land* and origin fixed effects. Pair fixed effects would absorb the cross-sectional IV. Column (2) is the delta-version PPML reduced form, replacing the 2016 stock exposure with the 2014–2016 predicted delta exposure. Column (3) replaces *Land* × year fixed effects with explicit controls for log GDP, log population, unemployment rate, log employment, manufacturing share, and log total *Land*-exports to the world. It ends in 2024 because the VGR der Länder regional-control aggregates for 2025 had not yet been released when the data were assembled; the main specification in Table 1, which absorbs these controls in θ_{it}, extends through 2025.

7.7 Consolidated robustness package

Appendix Table A1 subjects the preferred PPML reduced form to five further robustness checks: excluding the COVID years, excluding Eritrea, using export weights instead of export values, extending the exposure window to the 2014–2017 delta, and estimating five leave-one-origin-out specifications. None of the coefficients is statistically significant, and Appendix Table A2 shows that the first-stage F-statistic remains above the conventional threshold in every leave-one-origin-out sample. The null, therefore, survives changes in the sample period, origin composition, outcome variable, exposure definition, and origin exclusions.

7.8 Interpretation: an informative null

Taken together, the estimates provide no evidence of a positive export response to refugee-induced regional exposure in this setting and over this horizon. This is not a precise zero in the sense of a tightly identified small coefficient: the standard error of 0.0044 on the preferred reduced form is large relative to the point estimate of -0.0001, so small positive responses cannot be statistically ruled out. The estimates also do not support a sizeable positive response comparable to the large elasticities found in some cross-country aggregate studies. However, the magnitudes are not directly comparable because the treatment is measured as regional refugee exposure per 1,000 persons. The relevant claim is therefore not that migration does not affect trade, but that this particular design — a formal-allocation refugee shock, five origins, sixteen *Länder*, an unbalanced panel of 1,247 cell-year observations, and a nine-year post-shock window — detects no positive response. Because the first stage is strong, the pre-trend test provides no evidence of differential pre-shock trends, and the event study shows no visible break at 2015–2016. The null is robust across estimators, samples, outcomes, exposure definitions, and origin exclusions; the absence of a positive response is best read as an informative null rather than as a measurement failure.

8 Discussion and Conclusion

This paper assessed two contributions to the migration-trade literature and developed a research proposal that revisits the migrant-network–trade link in a contemporary German setting. The design exploits the *Königsteiner Schlüssel* as a plausibly exogenous allocation rule for the 2015/16 inflow of protection seekers from Afghanistan, Eritrea, Iraq, Iran, and Syria, observed at the *Land* $\times$ origin $\times$ year level across all sixteen *Länder* between 2010 and 2025. The preliminary analysis returns an informative null: the post-2016 origin-specific export response to predicted network exposure is statistically indistinguishable from zero, and this holds despite a strong first stage and across the full robustness package documented in Section 7.

Interpretation relative to Parsons and Vézina

Four structural differences help explain the contrast with the large, positive elasticities reported by Parsons and Vézina (2018). First, their post-shock horizon spans 15 years following the lifting of the embargo, whereas the present design spans 9. Second, Germany already maintained national-level exports to all five origins throughout the pre-period; unlike Vietnam in 1994, there is no zero-trade discontinuity, so comparable extensive-margin gains are less likely. Third, the 1975 Vietnamese boat-people cohort and the 2015/16 refugees in Germany are not interchangeable populations: the two cohorts differ in conflict context, average education at arrival, and legal pathways to the destination (Brücker et al., 2017). Fourth, Parsons and Vézina's effect is concentrated in differentiated goods (Rauch, 2001), a margin that this design cannot separately identify because the outcome is the total *Land*-level export value across all product categories. The design carries the same no-interference concern raised against Parsons and Vézina, since networks in one *Land* may contribute to exports recorded as departing from another.

Interpretation relative to Orefice, Rapoport, and Santoni

The positive bilateral elasticities reported by Orefice et al. (2025) are based on a design with a much larger cross-section of country pairs, pooled migrant types, and shift-share instruments, whose share-exogeneity assumption is scrutinised in Section 2, following Goldsmith-Pinkham et al. (2020). This design replaces historical settlement shares with a formula-based share that is less likely to reflect pre-existing origin-specific trade ties. The price is statistical power: identifying variation rests on $J = 5$ origins and 1,247 cell-year observations. The null, therefore, does not contradict their positive estimate; it suggests that, under a formal allocation rule and a smaller origin dimension, the short-run refugee-origin elasticity in this setting is small and imprecisely estimated. A further consideration is skill composition: Orefice et al. (2025) document in their Table 6 that the transaction-cost effect is stronger for tertiary-educated migrants, a margin on which the 2015/16 protection-seeker cohort plausibly differs from their pooled global sample.

Lag structure

Jaeger et al. (2018) argue that shift-share migration designs can blend short- and long-run effects when migration and outcomes evolve jointly. By construction, the proposed design isolates the short-run response: the allocation rule pins down a one-time shock, and the post-period covers only the nine years following it. Read through this lens, the null is informative about the short run specifically and does not speak to whether networks shape trade once firms have had more time to identify buyers, refugees to acquire language and labour-market attachment, and networks to deepen. Extending the post-period, however, is not a clean remedy. As noted in Section 5, the *Königsteiner Schlüssel* governs only the initial allocation, and the gap between formula-based and actual local stocks widens with the horizon as protection seekers relocate after asylum.

A natural extension

A natural extension would compare the elasticities of refugees and economic migrants within the same destination. The present design isolates a refugee shock, while Orefice et al. (2025) pool migrant types. A simple decomposition would write the bilateral elasticity as

$$\beta_{ij} = s_{ij}^{R}\beta^{R} + s_{ij}^{E}\beta^{E},$$

with s_{ij}^{R} and s_{ij}^{E} denoting type-specific population shares and β^{R} and β^{E} the corresponding elasticities. These components need not coincide given differences in time horizons, skill selection at arrival, and pre-arrival origin ties (Cortes, 2004; Grogger and Hanson, 2011). Identifying both would require two separate sources of exogenous variation and is beyond the proposed design. A parallel extension concerns the return-migration channel of Bahar et al. (2024), who document that returning refugees raise export performance in their origin countries through a knowledge-diffusion mechanism. The present design captures only refugees who remain in Germany, leaving unmeasured the export effects of eventual returnees.

Mechanism, scope, and future research

Linking the design to firm- and worker-level data — the IAB-BAMF-SOEP refugee survey, LIAB, or customs microdata — would address the firm-level limitation raised in Sections 2.1.3, 5 and 6, by recovering the buyer-seller and worker-firm interactions of Gould (1994), Rauch (2001), and Rauch and Trindade (2002), that *Land*-level exports cannot. Further extensions could test whether responses concentrate on differentiated goods or recover the sectoral knowledge-diffusion channel of Orefice et al. (2025) via the comparative-advantage construction of Costinot et al. (2012). The findings remain tightly delimited: under a formal institutional shift-share design, the migrant-trade nexus is not clearly visible in the short run at the *Land*-origin level.

Appendix

A.1 Consolidated Robustness Package

Table A1: Consolidated robustness package

	(1) Drop COVID	(2) Drop Eritrea	(3) Export weight	(4) Delta endpoint 2014–2017	(5) Leave-one-origin-out range
IV: predicted exposure (/1,000)	-0.0015	-0.0001	0.0055	-0.0001	[-0.0020, 0.0018]
	(0.0043)	(0.0045)	(0.0153)	(0.0076)	
Outcome	Exports	Exports	Weight	Exports	Exports
Estimator	fepois	fepois	fepois	fepois	fepois
Sample period	excl. 2020 & 2021	2010–2025	2010–2025	2010–2025	2010–2025
Fixed effects:					
Land × origin	Yes	Yes	Yes	Yes	Yes
Land × year	Yes	Yes	Yes	Yes	Yes
Origin × year	Yes	Yes	Yes	Yes	Yes
Observations	1,095	1,020	1,247	1,247	991–1,020
Clusters (pair)	80	64	80	80	64–80

Notes: Robust standard errors clustered at the *Land* × origin country level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Each column reproduces the preferred PPML reduced-form specification from Main Table 1, column (2), under the indicated robustness condition. Column (5) reports the range of point estimates across five regressions, each excluding one of the five origin countries; the full set of leave-one-origin-out estimates appears in Appendix Table A2. The instrument in column (4) is the 2014–2017 predicted delta exposure. None of the coefficients is statistically significant at conventional levels.

A.2 Leave-One-Origin-Out Detail

Table A2: Leave-one-origin-out detail

	Coefficient	SE	First-stage F	N
Drop Afghanistan	-0.0006	0.0045	29.1	995
Drop Eritrea	-0.0001	0.0045	20.7	1,020
Drop Iraq	0.0011	0.0040	21.7	991
Drop Iran	-0.0020	0.0057	18.0	991
Drop Syria	0.0018	0.0111	11.1	991

Notes: Robust standard errors clustered at the *Land* × origin country level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Each row reproduces the preferred PPML reduced-form specification from Main Table 1, column (2), excluding the indicated origin country. First-stage F-statistics remain above the conventional weak-instrument threshold of 10 in every specification. None of the coefficients is statistically significant at conventional levels.

A.3 PPML Event-Study Figure

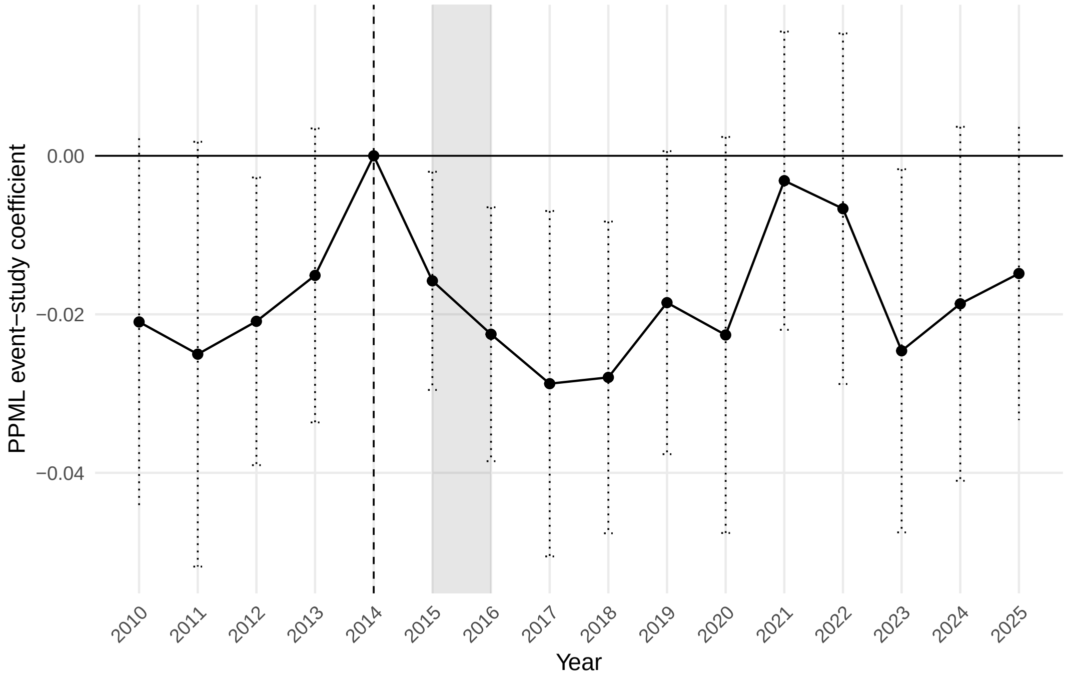

Figure A1: PPML event-study estimates

Notes: The figure plots year-specific PPML coefficients on the interaction between year indicators and the 2016 *Königstein*-predicted exposure measure, with 2014 as the reference year. Dotted whiskers indicate 95% confidence intervals constructed from cluster-robust standard errors at the *Land* $\times$ origin level. The shaded band indicates the 2015–2016 shock period. Most coefficients lie below zero, including in pre-shock years; this reflects the choice of 2014 as a single high reference year on the PPML scale rather than a differential post-shock effect. The pattern is consistent with the linear event study in Main Figure 1 and with the static null result in Main Table 1.

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Documentation of LLM Use

I used large language models as a support tool during the preparation of this paper. All substantive decisions, empirical interpretation, and final wording were reviewed and edited by me. In particular, I used LLM assistance for the following tasks:

1. **Grammar, wording, and character-limit management** I used LLMs throughout the writing process to improve grammar, wording, readability, and concision. Draft paragraphs were written or substantially revised by me first, then edited with LLM assistance; suggestions were reviewed selectively and adapted to fit the paper's argument and character limit.

2. **LaTeX formatting and bibliography support** I used LLMs for assistance with LaTeX commands, table formatting, cross-references, footnotes, appendix formatting, and the manual bibliography. This affected the formatting of tables, figures, the appendix, and the bibliography, but not the substantive interpretation of the results.

3. **Structuring sections and creating checklists** I used LLMs to brainstorm possible outlines, construct section-level checklists, and check whether each section addressed the assignment requirements. This was used mainly for the introduction, the literature assessment, the empirical strategy, and the discussion.

4. **Clarifying methods and concepts of assigned papers** I used LLMs to clarify econometric concepts such as shift-share instruments, PPML, control-function IV-style PPML, pre-trend tests, and extensive versus intensive margins. These outputs were checked against the original papers and course material before being rewritten in my own words, mainly for Sections 2 and 4.

5. **Brainstorming and developing the research proposal** I used LLMs in the early stages of the project to discuss possible research settings, mechanisms, and identification strategies for the research proposal. This support informed the development of the German refugee-allocation setting in Sections 3 and 4, but the final research question and empirical design were selected and adapted by me.

6. **Searching for and organising supporting sources and data** I used LLMs to help identify relevant datasets, data portals, and supporting literature for the research proposal, including German regional trade data, protection-seeker statistics, the *Königsteiner Schlüssel*, and possible firm- or worker-level extensions. The final sources used in Sections 4, 6, and 8 were verified and cited separately.

7. **Code review and empirical workflow support** I used LLMs to help clean and review R code for data construction, regressions, robustness checks, and event-study outputs. The final code was run and checked by me, and the resulting estimates are reported in Section 7 and the Appendix.

8. **Robustness checks supporting interpretation of the null result** I used LLMs to brainstorm possible robustness checks for the preferred PPML reduced-form specification and to discuss how to interpret the resulting null finding. The final robustness checks reported in Section 7 and the Appendix were selected and interpreted by me.